Exploring Early-Winter Snow Cover Change in Recent Decades Across Minnesota

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Background

- Across the Northern Hemisphere, temporal extent of snow cover has decreased as a result of climate change (Riggs & Hall, 2020)
- Snow cover extent plays an important role in mediating the Earth's energy balance and source of freshwater (Riggs & Hall, 2020)
- Decrease in snow cover adversely affects recreation and tourist-based economies
    - Many winter recreation-based areas are projected to lose most or all of their annual natural snow cover by 2071 (Mitterwallner et al., 2024)
- Broadly, a loss of snow cover brings both economic and environmental resource strain (Filho et al., 2024)

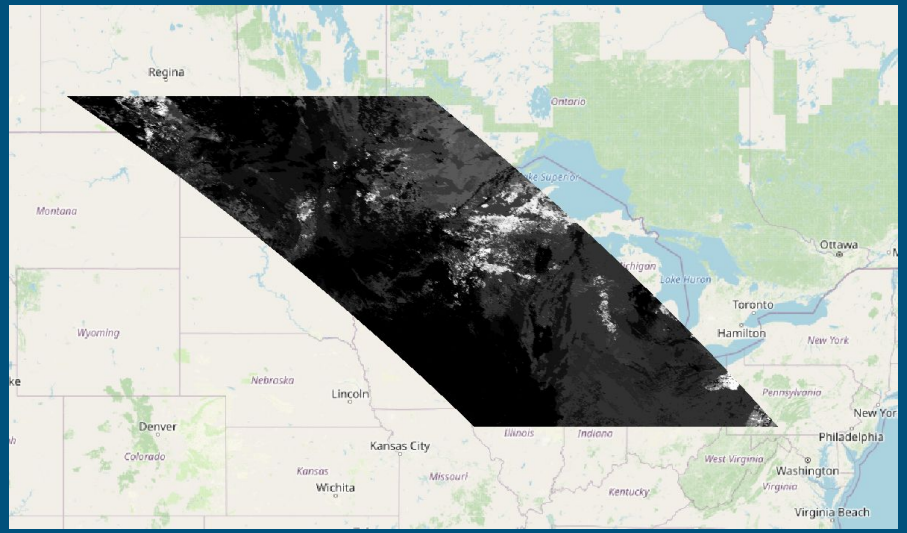
Research question

- For the first 28 days of December, how has Minnesota's snow cover changed from 2000 to 2024?

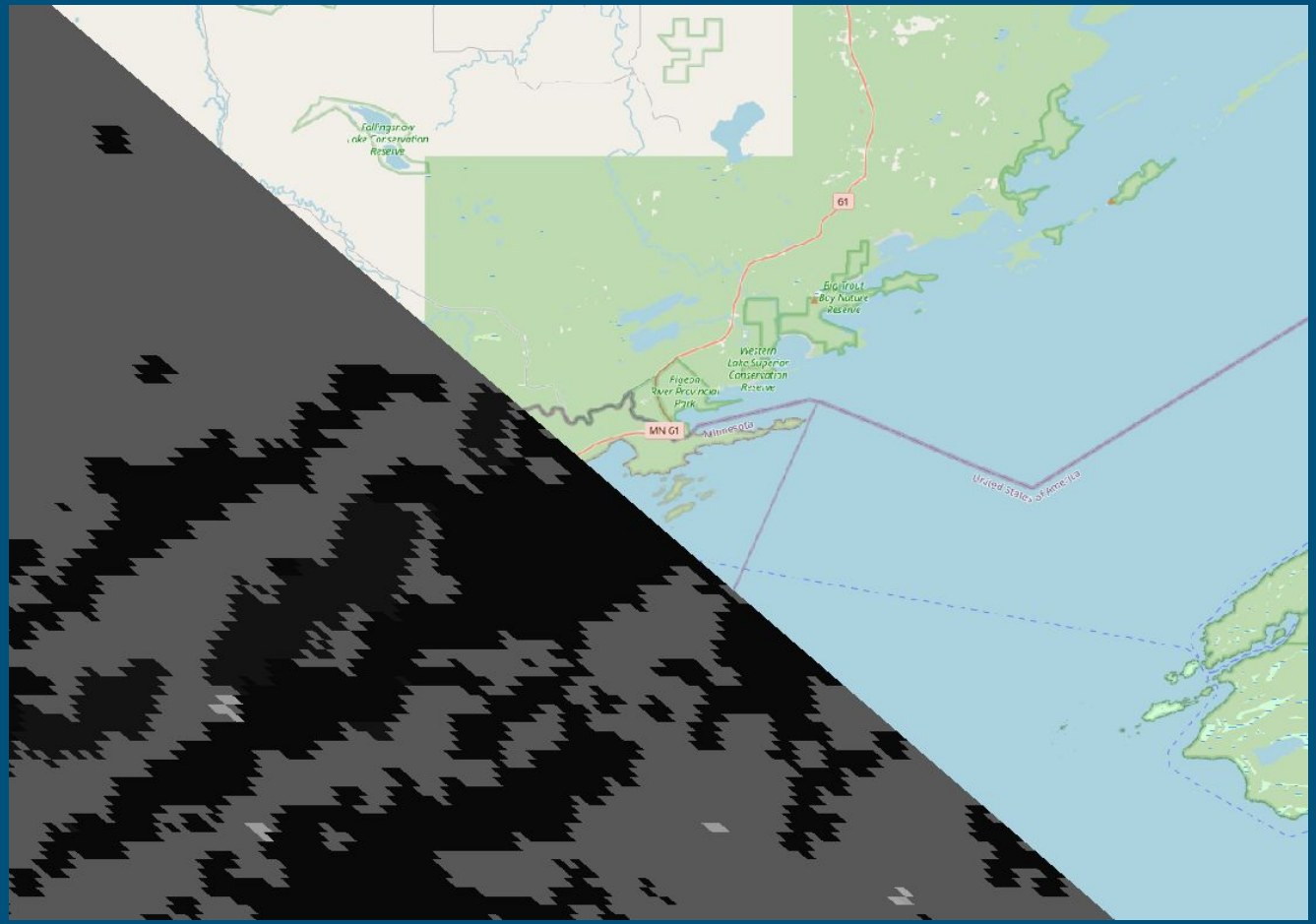
- Expectation is that as latitude decreases, slope of regression line (at the pixel level) becomes more negative (losing snow cover faster)
  - Additionally, expected that as year increases, proportion of MN covered by snow decreases

Data

- MODIS CGF Snow Cover Daily Global 500m Grid product
- Tile h11v4 (covers almost all of MN) from daily NDSI snow cover product
- Spatial resolution of 500 m, each tile is 10 degrees by 10 degrees
- Daily imagery



Values
NDSI snow cover values and data flags values, stored as 8-bit unsigned integers.
0 - 100: NDSI snow cover (no snow to completely snow covered)
200: missing data
201: no decision
211: night
237: inland water
239: ocean
250: cloud
254: detector saturated
255: fill



Far northeastern part of MN not included

Accessing data

- earthaccess library in Python
- Had issues with MODIStsp in R
- Files are in HDF format
- 700 HDF files – first 28 days of December, 2000 through 2004

```
import earthaccess

earthaccess.login()

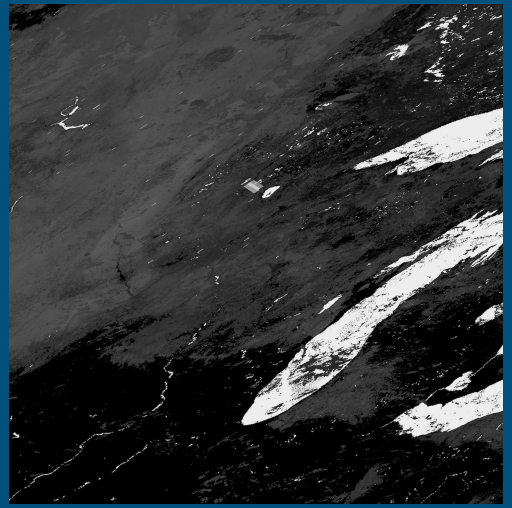
results = earthaccess.search_data(
    short_name= "MOD10A1F",
    temporal = ("2024-12-01","2024-12-28"),
    point = (-94.35772, 45.64187)
    )

print(len(results))

earthaccess.download(results, local_path: "/Users/sspisak23/Documents/GEOG_352/MODIS")
```

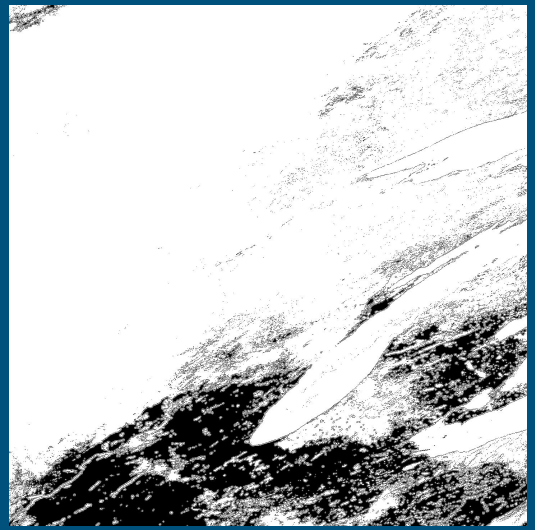
Converting files to geotiffs

- terra library in R
- Create list of files from all HDF files
- Loop over the list, create raster object from each file, save it as a geotiff, and save it in yearly and weekly folder based on its julian date
- 700 geotiffs



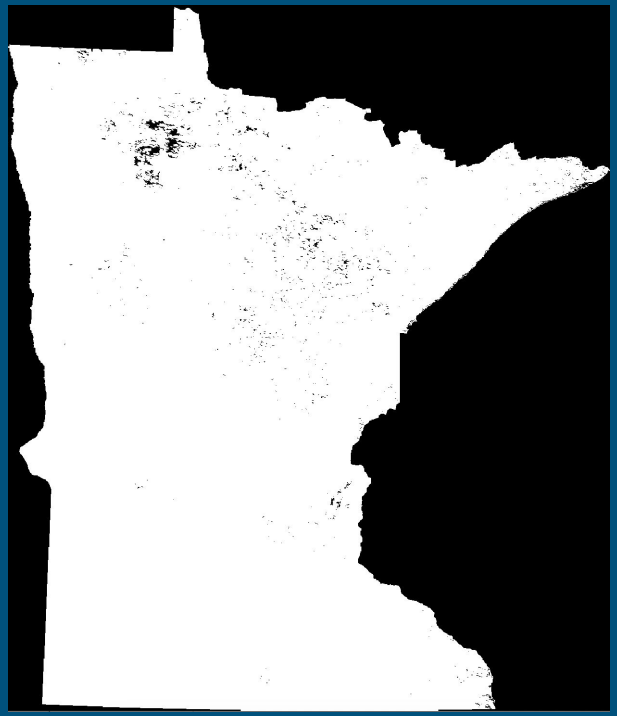
Creating weekly averages

- Loop over folders and create list of files from each weekly folder of each year
- Create list of raster objects
- Create raster stack from list
- Calculate weekly average from raster stack and save in folders for week 1, 2, etc.
- 100 geotiffs



Reproject, cropping, and trimming

- Loop over files in each folder
- Create raster object
- Define sinusoidal projection and project to NAD83 UTM Zone 15N
- Crop to MN boundary using shapefile of MN using MNDOT’s state boundaries shapefile
- Trim raster to remove rows/columns that are all NA
- Save in folders based on week



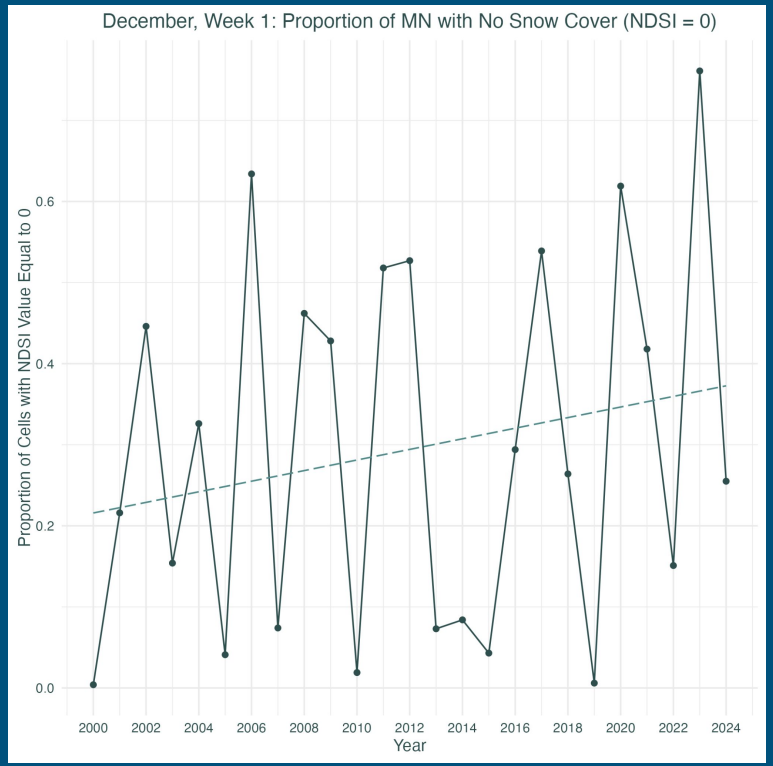
Calculate proportions

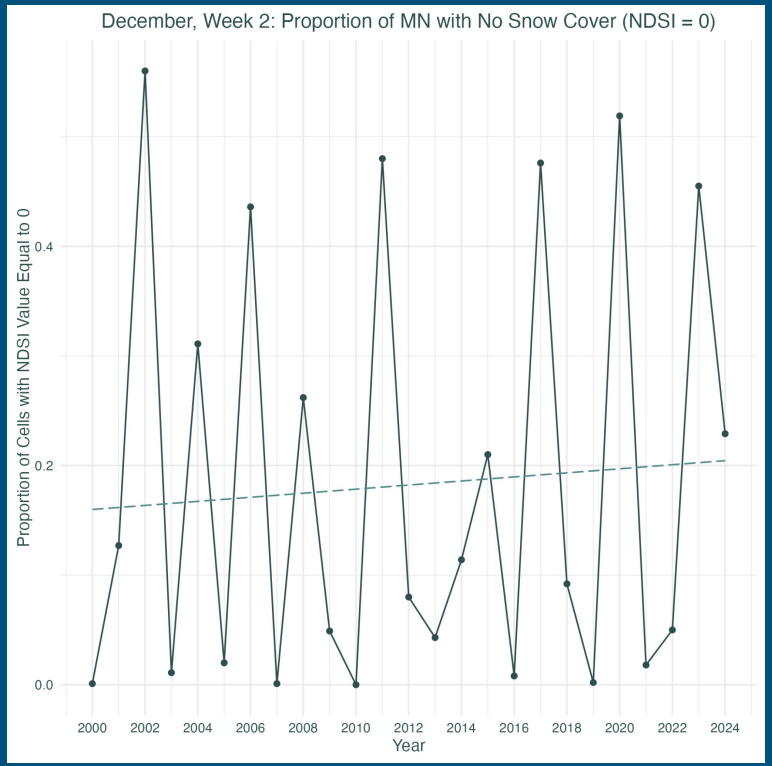
- Loop over files and create raster objects
- Get count of all non-NA cells in raster
- Get count of all cells with value equal to 0
- Calculate proportion of cells where value equals 0
- Save in spreadsheet

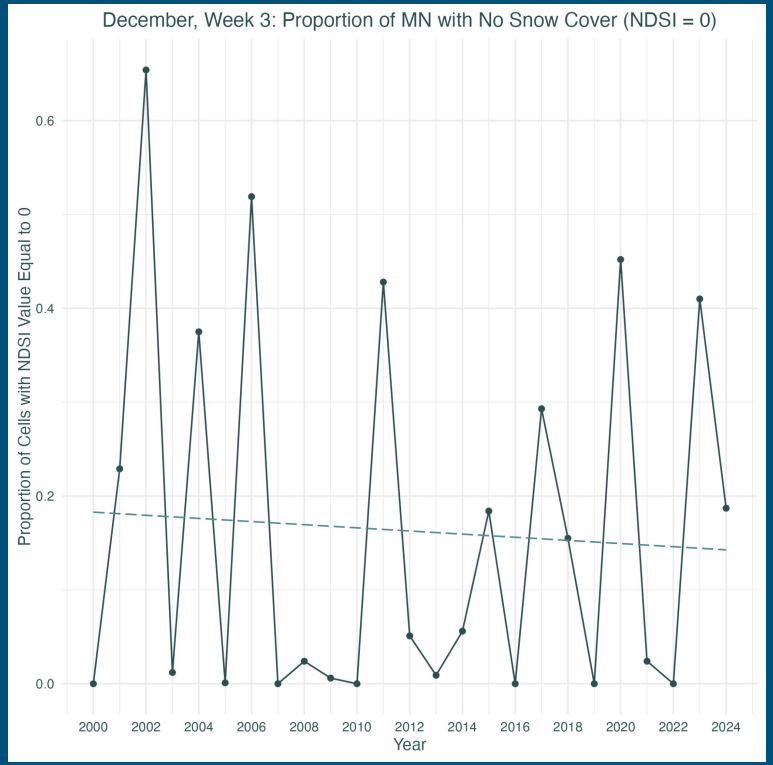
Create rasters showing change over time

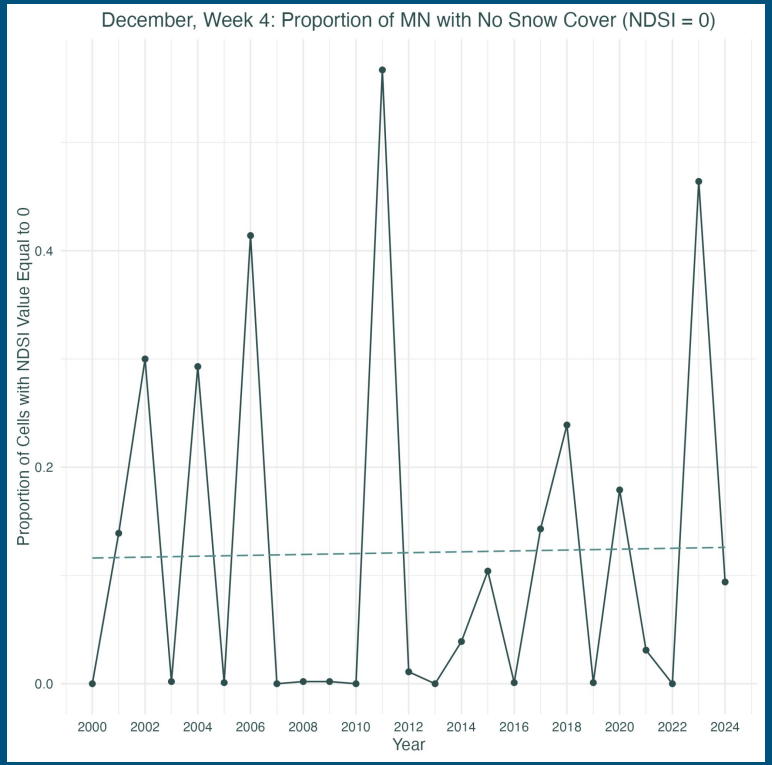
- Create raster objects from list of files
- Create stack of raster objects
- Resample to 1 km resolution
- Used AI to create code for slope calculation function

- Save raster as data frame and visualize with ggplot2

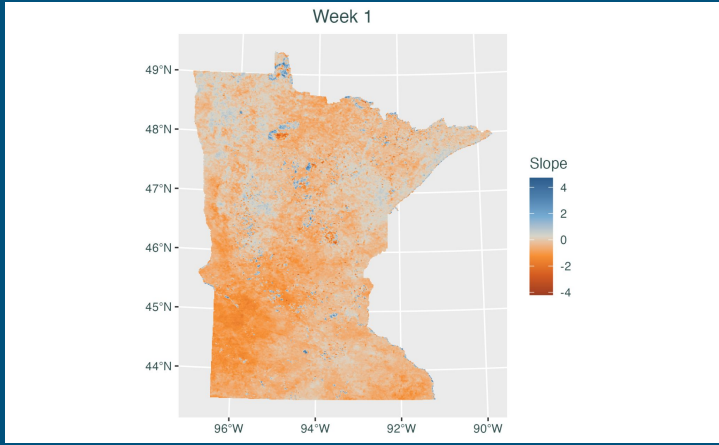

December, Week 1: Proportion of MN with No Snow Cover (NDSI = 0)
Proportion of Cells with NDSI Value Equal to 0
0.0
0.2
0.4
0.6
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
Year

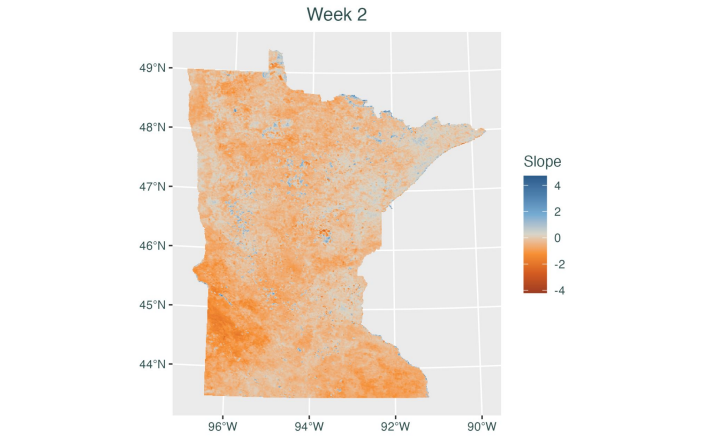

December, Week 2: Proportion of MN with No Snow Cover (NDSI = 0)
Proportion of Cells with NDSI Value Equal to 0
0.0
0.2
0.4
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
Year

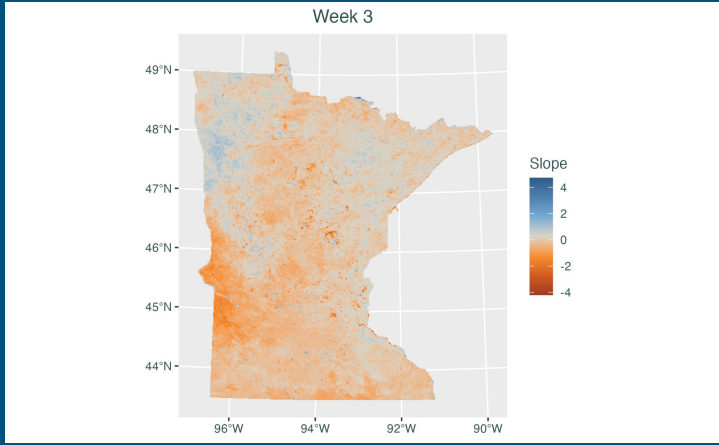

December, Week 3: Proportion of MN with No Snow Cover (NDSI = 0)
Proportion of Cells with NDSI Value Equal to 0
0.0
0.2
0.4
0.6
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
Year

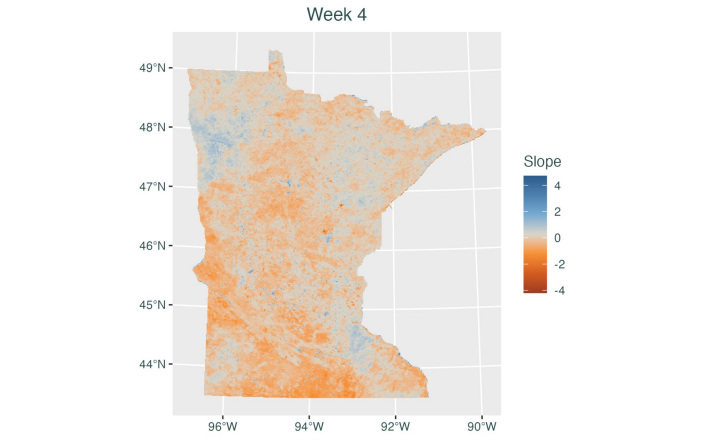

December, Week 4: Proportion of MN with No Snow Cover (NDSI = 0)
Proportion of Cells with NDSI Value Equal to 0
0.0
0.2
0.4
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
Year

Maps depicting slope (NDSI change over years) at pixel level









Limitations

- Not handling non-NDSI values before creating averages

- Cloud-gap filled means data might not actually be from that day

Literature Referenced

Filho, W., Dinis, M., Nagy, G., Fracassi, U., & Aina, Y. (2024). A ticket to where? Dwindling snow cover impacts the winter tourism sector as a consequence of climate change. *Journal of Environmental Management*, *356*.

Mitterwallner, V., Steinbauer, M., Mathes, G., & Walentowitz, A. (2024). Global reduction of snow cover in ski areas under climate change. *PloS One*, *19*(3).

Riggs, G., & Hall, D. (2020). Continuity of MODIS and VIIRS Snow Cover Extent Data Products for Development of an Earth Science Data Record. *Remote Sensing*.